

# DHANVIN S

College Park, MD | dhanvinsreddy@gmail.com | (240) 927-9622 | linkedin.com/in/dhanvin-s-8ab99b220 | github.com/DhanvinS

Machine Learning Engineer | AI Evaluation & LLM Systems | Benchmarking, Regression Detection & Model Quality

## EVALUATION SYSTEMS & LLM QUALITY

### Multimodal VLM Evaluation & Failure Analysis

PyTorch · Hugging Face · Vision-Language Models · MMMU

- Built a reusable evaluation framework for vision-language models over ~4,000 MMMU image-question pairs, designed so new models and prompting strategies drop in without touching the pipeline, so comparisons stay consistent across experiments and users; cut manual review effort ~90%.
- Evaluated at capability level rather than aggregate: 68% overall accuracy masked a ~20 point gap between natural image understanding (77%) and charts and tables (57%), with OCR and numerical reasoning similarly weak and 15 to 18 point drops on questions with dense embedded text.
- Built a failure taxonomy and manually reviewed ~600 incorrect predictions, attributing ~60% to visual grounding, OCR, or chart interpretation, reframing many apparent reasoning failures as information-extraction failures. Structured visual reasoning prompts lifted accuracy to 71% with ~5 points on charts, but left OCR slices flat, isolating perception as the bottleneck.

### LLM & Agent Evaluation Harness

LangChain · Hugging Face · Anthropic & OpenAI APIs · FastAPI

- Built the evaluation system for agentic LLM workflows: 15+ metrics over 10K+ curated cases scoring response quality, groundedness, and hallucination rate, combining LLM-as-judge with structured metrics and semantic clustering of failures to surface modes the metric suite was not yet measuring.
- Added experiment logging and regression tests across model, prompt, and retrieval configurations so any change could be attributed and caught before shipping, and served it through FastAPI in containers so runs stayed reproducible and comparable across sessions and machines.

### ML Insight, Automated Model Quality & Regression Analysis Platform

Python · XGBoost · FastAPI · PostgreSQL · AWS

- Built infrastructure evaluating model behaviour across 100+ dynamic data slices with drift detection, calibration, SHAP attribution, and automated root-cause reporting. Across 1M+ predictions it surfaced 17 undetected high-error segments, cut slice-level error 23%, improved calibration from ECE 0.081 to 0.027, and cut debugging from hours to under 15 minutes.

### Adversarial Benchmarking & Measurement Validity

PromptGuard · HDBSCAN · UMAP · Llama 3.x · Kaggle

- Built adversarial evaluation pitting adaptive attacker agents against defender classifiers across 8 harm categories with per-round Elo ratings, plus open-set evaluation measuring generalisation to held-out families at 0.87 AUROC, the case where the benchmark contains nothing like the input.
- Held first place for a month on the public board of an agent security competition hosted by OpenAI, Google, and IEEE; the result did not transfer to the hidden private board, a measurement-validity failure I traced to optimising a visible proxy over the true objective. Separately found a label-leakage bug inflating my own benchmark and rebuilt a saturated eval with hard negatives.

## EXPERIENCE

### Machine Learning Engineer, Accenture | Bengaluru, India

Aug 2023 – Jul 2025

Python, PyTorch, SQL, Spark, AWS, Docker, Kubernetes, MLflow, Git, CI/CD

- Owned model quality end to end over 3M+ events per day: evaluation design, calibration, and segment-level analysis across 10+ cohorts that surfaced gaps aggregate metrics concealed, with holdout experiments, hypothesis testing, and confidence intervals to establish whether a change actually helped.
- Built scalable infrastructure supporting training, benchmarking, and deployment: distributed Spark processing, MLflow tracking and model versioning across 20+ iterations, containerised serving at 99.9% availability, and CI/CD with automated testing that cut release time 32%.
- Built monitoring for drift (KS/PSI), calibration, and data-quality failure wired to trigger retraining, and triaged production regressions through root-cause analysis across data, features, infrastructure, and model behaviour.
- Wrote production Python with unit testing and code review, and made the evaluation and monitoring tooling usable by others: documented methodology, assumptions, and API contracts so engineers, analysts, and product teams could consume the models and read the quality signals without me, cutting onboarding from one week to two days.

### AI / LLM Intern, PES University C-IoT | Bengaluru, India

Mar 2022 – Aug 2022

- Fine-tuned RoBERTa for content quality classification over 50K+ samples to 88% F1, and built automated safety-evaluation pipelines across 2K+ hard adversarial examples that lifted unsafe-content recall 18%.

## ADDITIONAL RESEARCH & CONTRIBUTIONS

### Learned Timestep Sampling in Diffusion Models

PyTorch · UNet · paper in preparation

- Derived and implemented a novel training method from first principles: a learnable timestep distribution under importance weighting and entropy and coverage constraints, evaluated over the full training trajectory (lowest FID-AUC of all schedules) rather than a single checkpoint, reaching FID 136.10 in 10K steps where uniform sampling does not in 100K.
- Open-source: merged 14 pull requests across 3 repositories including API tooling and a Python data-validation library, working inside existing conventions and through maintainer review on implementation quality and test coverage.

## TECHNICAL SKILLS

**Evaluation & Analysis:** Eval and benchmark design (MMMU and custom), multimodal and capability-level evaluation, failure taxonomy and error attribution, LLM-as-judge, groundedness and hallucination scoring, open-set evaluation, regression testing, slice-based analysis, calibration (ECE), drift detection (KS/PSI), SHAP, A/B testing

**ML & LLM:** PyTorch, TensorFlow, Hugging Face Transformers, vision-language models, multimodal inference, LangChain, FAISS, supervised fine-tuning, LoRA/QLoRA/PEFT, distillation, RAG, agentic workflows, prompt engineering, XGBoost, sequence models, GNNs, diffusion and generative models

**Engineering:** Python, C++17/20 (STL, RAII, atomics and memory ordering, lock-free structures, multithreading), Java, SQL, Bash, object-oriented design, Git, unit testing, CI/CD, code review, FastAPI, Docker, Kubernetes, Linux

**Scale & Cloud:** Distributed data processing (Spark, Ray), AWS (SageMaker, Lambda, S3), MLflow, Optuna, large-scale dataset curation and versioning, model serving and monitoring

## EDUCATION

University of Maryland, College Park — M.Sc. Applied Machine Learning | GPA 3.9/4.0

May 2027

PES University, Bengaluru — B.Tech. Computer Science & Engineering | GPA 8.5/10

May 2024